

Windows Administration Points

1 Windows Server Fundamentals

1. Windows Server is used to manage network resources and users.
2. Windows Server provides centralized administration.
3. Windows Server supports Active Directory services.
4. Windows Server manages users, computers, and policies.
5. Windows Server supports file and printer sharing.
6. Windows Server supports virtualization using Hyper-V.
7. Windows Server improves security and management.
8. Windows Server supports remote administration.
9. Windows Server uses NTFS file system for security.

2 User and Group Management

1. Users are created to allow login access.
2. Groups are used to assign permissions easily.
3. Security groups are used for permission assignment.
4. Distribution groups are used for email distribution.
5. User accounts can be disabled to prevent login.
6. Password policies enforce password complexity.
7. Account lockout policy prevents unauthorized access.
8. User profiles store user environment settings.

3 File System and Permissions

1. NTFS provides file and folder security.
2. Permissions control access to files and folders.
3. Basic permissions include Read, Write, Modify, Full Control.
4. Inheritance allows permissions to flow from parent folder.
5. Share permissions apply over the network.
6. NTFS permissions apply locally and over network.
7. Effective permission is the combination of NTFS and Share permissions.
8. File compression reduces storage usage.
9. Encryption protects sensitive files.

4 Group Policy

1. Group Policy is used to manage user and computer settings.
2. GPO stands for Group Policy Object.
3. GPO can be applied to Site, Domain, and OU.

4. Group Policy can enforce security settings.
5. Software deployment can be done using Group Policy.
6. Login scripts can be assigned using Group Policy.
7. gpupdate command updates Group Policy.
8. Group Policy improves centralized management.

5 Windows Server Roles

1. Roles provide specific server functions.
2. Active Directory Domain Services manages domain.
3. DNS Server resolves names to IP addresses.
4. DHCP Server assigns IP addresses automatically.
5. File Server provides file sharing services.
6. Print Server manages printers.
7. Web Server (IIS) hosts websites.
8. Hyper-V provides virtualization services.
9. Windows Deployment Services installs OS over network.

6 Disk Management and Storage

1. Disk Management is used to manage disks and partitions.
2. Basic disks use primary and extended partitions.
3. Dynamic disks support advanced volumes.
4. RAID improves performance and fault tolerance.
5. RAID 0 improves performance but no fault tolerance.
6. RAID 1 provides mirroring and fault tolerance.
7. RAID 5 provides parity and fault tolerance.
8. Storage Spaces provides storage virtualization.

7 Server Monitoring and Maintenance

1. Task Manager monitors running processes.
2. Resource Monitor shows CPU, memory, disk usage.
3. Event Viewer logs system events.
4. Performance Monitor tracks system performance.
5. Windows Update installs security updates.
6. Services.msc manages background services.
7. Proper monitoring improves server performance.

8 Backup and Recovery

1. Backup protects data from loss.

2. Windows Server Backup is used for backup.
3. Full backup backs up entire system.
4. Incremental backup backs up changed data only.
5. System State backup backs up AD and system files.
6. Restore recovers data from backup.
7. Backup is important for disaster recovery.

Windows Server Security

1. Firewall protects system from unauthorized access.
2. Antivirus protects against malware.
3. User Account Control (UAC) improves security.
4. BitLocker encrypts disk drives.
5. Security policies control user rights.
6. Regular updates improve security.
7. Strong passwords improve system protection.